



EP

Engineering & Planning

Read the game, define the work, plan the season.

13 cards EP-1 — EP-13

WHAT BELONGS IN THIS FOLDER

EP-1	Team Identity: Name, Look & Motto	•	
EP-2	Read the Game, Build the Wall Poster	•	
EP-3	Robot Requirements List	••	
EP-4	Design Sketches & Pick a Concept	••	
EP-5	Set Up the Engineering Notebook	•	
EP-6	Tape Out the Practice Field	•	
EP-7	Shop Safety & Tool Check-Off	•	
EP-8	Scouting & Alliance Strategy	••	
EP-9	Drive Team: Roles, Calls & Practice	••	ROBOT + FIELD
EP-10	Pit Setup & Event Day Run Sheet	•	
EP-11	Judging: Portfolio & the Five-Minute Interview	•••	
EP-12	Post-Event Debrief: What Mk-2 Must Fix	••	
EP-13	Build the BIOBUZZ Field Elements	••	

Taking a sheet? Move your chip to CLAIMED on the board and write your names next to it.

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CD

CAD & 3D Printing

Model it in Onshape, then make it real.

9 cards CD-1 – CD-9

WHAT BELONGS IN THIS FOLDER

CD-1	Onshape Setup & Your First Part	•
CD-2	Measure & Model a Real Robot Part	••
CD-3	Design a Pollen Intake in CAD	•••
CD-4	Design a Pollen Outtake in CAD	•••
CD-5	Build the Robot in CAD: Assembly & Fit Check	•••
CD-6	From CAD to Printed Part	••
CD-7	Design a Printed Part That Solves a Real Problem	•••
CD-8	Design the Odometry Mounting Adapter	•••
CD-9	Fit the Starter Bot to Our Chassis in CAD	•••

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PR

Programming

Make the robot do what you tell it.

21 cards PR-1 – PR-21

WHAT BELONGS IN THIS FOLDER

PR-1	Get Connected: Control Hub & Driver Hub	•	BOARD / ROBOT
PR-2	Build the Robot Configuration File	•	BOARD / ROBOT
PR-3	First TeleOp: Make It Drive (Mecanum)	•	ROBOT
PR-4	Telemetry: See What the Robot Is Thinking	••	BOARD / ROBOT
PR-5	Driver Tuning: Slow Mode & Smooth Control	••	ROBOT + FIELD
PR-6	First Autonomous: Time-Based Movement	••	ROBOT + FIELD
PR-7	Encoder Autonomous: Drive an Exact Distance	•••	ROBOT + FIELD
PR-8	Intake Control: Map a Mechanism to a Button	••	ROBOT
PR-9	Practice Without the Robot: FTC SIM	•	
PR-10	Touch Sensor: The First Decision	•	BOARD
PR-11	Color Sensor: Teach It to See Pollen	••	BOARD
PR-12	Potentiometer & Servo: Analog In, Position Out	••	BOARD
PR-13	Bench to Robot: Move Your Code Over	•••	ROBOT
PR-14	Field-Centric Drive with the IMU	•••	ROBOT + FIELD
PR-15	Read the Pinpoint: Configure & Prove It	••	ROBOT
PR-16	Pedro Pathing: Install & Tune the Localizer	•••	ROBOT + FIELD
PR-17	Pedro Pathing: Tune the Follower	•••	ROBOT + FIELD
PR-18	Starter Bot Configuration: Six Motors, Two Hubs	••	ROBOT
PR-19	Run the Starter Bot TeleOp on Mecanum	•••	ROBOT + FIELD
PR-20	Flywheel Tuning: Make the Shot Repeatable	•••	ROBOT + FIELD
PR-21	Starter Bot Autonomous: Score the Pre-Loads	•••	ROBOT + FIELD



BA

Building & Assembly

Cut, bolt, wire, and make it hold together.

18 cards BA-1 – BA-18

WHAT BELONGS IN THIS FOLDER

BA-1	Inventory the Kit & Set Up the Shop	.	
BA-2	Mecanum Chassis: Build the Frame	.	ROBOT
BA-3	Mecanum Drive: Gearboxes, Motors & Wheels	..	ROBOT
BA-4	Electronics: Mount & Wire the Control Hub	..	ROBOT
BA-5	Design & Build an Intake	...	ROBOT
BA-6	Battery Management Station	.	
BA-7	Scoring Mechanism Prototype (cardboard first)	...	ROBOT
BA-8	Robot Cart & Pit Kit	.	ROBOT
BA-9	Assemble a Programming Board	.	BOARD
BA-10	Install the Pinpoint Odometry Pods	..	ROBOT
BA-11	Starter Bot: Parts Check & Cut List	..	
BA-12	Starter Bot: Launcher Floor	..	
BA-13	Starter Bot: Intake	..	
BA-14	Starter Bot: Launcher Flywheel & Feeder	...	
BA-15	Starter Bot: Pollen Pusher	.	
BA-16	Mount the Starter Bot on Our Mecanum Chassis	...	ROBOT
BA-17	Add the Expansion Hub: Power, Comms & Six Motors	...	ROBOT
BA-18	Inspection Dry Run: Make It Legal	..	ROBOT

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MG

Merch & Give-Away

Things we hand people so they remember us.

6 cards MG-1 — MG-6

WHAT BELONGS IN THIS FOLDER

MG-1	Give-Away Plan & Budget	•
MG-2	Design the Give-Away Art	••
MG-3	Produce the Give-Aways (assembly line)	•
MG-4	Team Card / Business Card	••
MG-5	3D-Printed Give-Away	•••
MG-6	Design the Award Stickers	••

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TS

Team T-Shirts

What the team looks like at an event.

5 cards TS-1 – TS-5

WHAT BELONGS IN THIS FOLDER

TS-1	Shirt Design Concepts	•
TS-2	Vote & Make the Art Print-Ready	••
TS-3	Sizing Survey & Order Sheet	•
TS-4	Price the Shirts: Blanks + Transfers	••
TS-5	Press the Team Shirts	••

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MF

Marketing & Fundraising

Find the money and tell the story.

7 cards MF-1 – MF-7

WHAT BELONGS IN THIS FOLDER

MF-1	Build the Sponsor Prospect List	•	
MF-2	Sponsorship One-Pager	••	
MF-3	Write the Outreach Emails & Letters	••	
MF-4	Social Media Plan & First Posts	•	
MF-5	Team Page Content for the Herobotix Website	••	
MF-6	Plan a Demo / Outreach Event	••	ROBOT
MF-7	Grants & Team Funding Applications	•••	

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IN PROGRESS

Started but not finished.

A sheet lives here when the meeting ended before the work did. Write on the back where you stopped and what the next crew needs to know. These get picked up before anyone starts something new.



DONE & STAMPED

Every “Done when” box is true.

Stamped sheets wait here until they are logged and taped into the engineering notebook. A stamped sheet is already a notebook entry — it carries the checklist, the names and the date.